

A scientific figure in three attempts

Same synthetic dataset. Three fresh tasks.

1 Naïve prompt

2 Advanced prompt

3 Plan mode, review and execution

The dataset and the question

How does Compound_X affect CD8 activation with and without stimulation?

10 simulated donors
4 conditions per donor
2 technical aliquots per culture

80 measurements
40 culture averages
10 paired donors

data.csv and study_context.md are identical in every round.
All measurements and the compound are fictional.

Setup · 3 minutes

Round 1: the naïve prompt

“Use data.csv and study_context.md to make a scientific figure showing how Compound_X affects CD8 T-cell activation with and without stimulation. Save the figure and the files requested in study_context.md in outputs/.”

Open round_1_naive in a fresh task. Paste PROMPT.txt.

Inspect the first result and preserve it for comparison.

Round 2: the advanced prompt

The brief makes the scientific decisions explicit

Average technical aliquots within each culture.
Pair treatments by donor and separate stimulation groups.
Show individual donors with clear units and a full caption.
Save the plotted values, code and regeneration command.
Run the code and inspect the saved outputs.

Open round_2_advanced in a fresh task. Paste PROMPT.txt.

Round 3: Plan mode

Same advanced task, with a review before execution

Open round_3_plan in a fresh task and select Plan mode.
Paste PROMPT.txt and read the proposed approach.
Check the replicate unit, aggregation, pairing and units.
Request corrections, then approve the reviewed plan.
Start execution and inspect the figure and saved table.

Keep the plan and record any decisions you changed.

Round 3 · 3 minutes to review, 7 minutes to execute

Comparison of the three figures

Which result can you verify?

Trace one plotted value to its two technical measurements.
Check that every condition includes all 10 donors.
Check pairing, units and the synthetic-data caption.
Open the code and test its regeneration command.

Use REVIEW_SHEET.md and record missing evidence.
What did reviewing the plan change?